

CHEBOYGAN, MI, USA

WMO: 720321

Lat: 45.654N

Lon: 84.519W

Elev: 635

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 04893

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-5.8	0.1	-11.5	3.0	-4.8	-6.2	4.0	1.6	26.4	21.3	24.3	23.2	3.1	220	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	85.0	69.5	81.9	68.2	79.3	66.8	72.4	80.7	70.8	78.4	69.2	76.0	10.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.1	113.5	76.8	68.3	106.3	74.9	66.4	99.4	73.2	36.5	79.8	35.0	79.0	33.6	76.8	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.6	18.9	16.9	DB	-14.7	90.6	8.1	2.6	-20.5	92.5	-25.2	94.0	-29.8	95.5	-35.6	97.4	(4)
(5)			WB	-15.1	76.0	8.0	1.1	-20.9	76.8	-25.5	77.5	-30.0	78.1	-35.8	78.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	19.8	19.6	28.0	39.5	51.3	60.9	66.9	65.4	59.0	47.6	36.4	26.6		
	DBStd	18.69	10.76	10.37	10.93	7.85	7.60	6.45	5.55	5.53	7.36	8.17	8.16	8.64		
	HDD50	4171	937	852	685	327	73	2	0	0	8	147	414	726		
	HDD65	8080	1402	1272	1147	766	430	158	43	61	209	543	858	1191		
	CDD50	1817	0	0	4	11	114	327	525	479	278	73	6	0		
	CDD65	250	0	0	0	0	6	34	103	74	29	4	0	0		
	CDH74	2219	0	0	7	9	118	338	933	588	202	24	0	0		
	CDH80	483	0	0	2	2	27	78	223	114	35	2	0	0		
(14) Wind	WSAvg	6.9	7.4	8.1	7.5	7.9	6.5	5.7	5.6	5.7	5.9	7.2	7.6	7.8		
(15) Precipitation	PrecAvg	29.3	1.7	1.3	1.7	2.4	2.7	2.6	3.0	3.2	3.7	2.7	2.5	2.0		
	PrecMax	40.5	3.9	4.1	5.2	6.1	7.3	7.5	7.7	7.6	9.4	6.0	5.4	5.7		
	PrecMin	13.3	0.3	0.2	0.1	0.3	0.3	0.4	0.3	0.2	0.3	0.3	0.4	0.3		
	PrecStd	5.5	0.8	0.8	1.2	1.3	1.3	1.4	1.7	1.7	2.0	1.4	1.2	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.8	47.3	62.6	72.4	84.0	86.8	90.0	88.0	85.1	77.5	66.3	52.9		
		MCWB	41.3	42.7	52.8	56.3	64.9	70.0	72.2	71.7	70.6	63.6	55.3	50.7		
	2%	DB	39.9	39.4	53.2	62.9	77.0	81.9	85.5	82.8	79.6	71.5	57.0	43.5		
		MCWB	36.5	36.1	46.7	51.0	62.2	67.9	70.0	69.4	66.7	62.8	50.4	40.0		
	5%	DB	36.6	36.2	46.4	56.6	71.7	78.0	82.0	80.2	75.5	64.7	52.4	40.4		
		MCWB	34.6	33.5	40.5	46.4	58.8	65.7	68.6	67.7	65.6	58.8	47.5	38.2		
	10%	DB	34.0	33.7	42.5	52.1	66.0	73.4	79.3	77.1	72.4	60.9	48.3	37.4		
		MCWB	32.8	31.5	37.4	43.8	56.5	62.9	67.2	66.0	64.1	55.2	44.6	35.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.1	43.7	55.2	58.9	68.5	73.1	75.2	74.1	72.5	68.0	56.6	50.4		
		MCDB	44.5	47.5	60.8	69.7	77.9	81.8	84.8	82.9	80.5	72.9	63.4	52.5		
	2%	WB	37.3	36.8	47.1	52.1	65.1	70.0	72.8	72.0	69.8	63.5	51.5	41.4		
		MCDB	39.1	39.3	52.2	60.5	73.5	79.0	81.1	79.7	75.8	68.2	55.4	42.9		
	5%	WB	34.6	33.8	41.4	48.0	61.2	67.6	70.9	70.2	67.8	59.6	48.0	38.4		
		MCDB	36.3	35.6	46.3	55.3	69.7	75.5	78.9	76.7	74.1	64.9	51.8	40.0		
	10%	WB	32.6	31.7	37.7	44.9	57.6	65.2	69.1	68.5	65.6	55.7	45.1	35.8		
		MCDB	33.9	33.2	41.4	50.7	64.7	71.9	76.6	74.5	71.0	60.5	47.7	37.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.9	15.1	17.2	17.7	21.8	21.3	21.0	19.9	19.7	15.5	12.2	10.2		
		MCDBR	12.7	14.8	22.7	27.1	30.5	25.9	24.2	22.9	23.0	22.8	19.8	13.3		
		MCWBR	10.7	12.5	16.0	16.6	16.9	13.8	12.1	11.7	12.5	14.7	14.2	11.8		
	5% WB	MCDBR	12.1	14.5	21.4	25.2	26.3	22.8	20.9	19.1	20.2	20.7	18.2	12.8		
		MCWBR	10.8	12.9	15.7	16.8	16.6	13.4	11.5	11.0	12.7	14.5	14.2	11.8		
(40) Clear-Sky Solar Irradiance	taub		0.301	0.293	0.317	0.358	0.376	0.395	0.392	0.384	0.369	0.335	0.303	0.280		
	taud		2.298	2.337	2.395	2.338	2.364	2.343	2.371	2.389	2.421	2.498	2.519	2.446		
	Ebn at Noon		252	282	290	285	283	277	276	274	269	264	255	252		
	Edh at Noon		25	29	32	37	38	39	37	35	31	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		362	656	1109	1461	1793	1945	1956	1660	1261	733	387	268		
	RadStd		40	66	115	153	151	140	77	103	103	85	54	26		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page